

Lab 4: Interpretations and Discussion

The numbers behind each figure, the paragraph that interprets it, and the Discussion that answers the question

DR. SATADISHA SAHA BHOWMICK · FALL 2026 · WEEK 06 — WEEK OF MONDAY 2026-09-28, IN SECTION

DELIVERABLES — DUE AT THE END OF YOUR LAB SECTION, ON BRIGHTSPACE

- ☐ Lab-4_FIRSTNAME_LASTNAME.Rmd/.qmd: the starter sheet, renamed, with the three parts completed as far as you got and the generative AI disclosure filled in
- ☐ Lab-4_FIRSTNAME_LASTNAME.pdf: the same sheet, knitted to PDF
- ☐ the data files the sheet reads: nashville_daily.rds and your own dataset

Report 1 (exploratory data analysis and visualization) is due on Friday 2026-10-09. In Lab 3 you drew its figures and wrote a caption and a paragraph for each. This lab is about the writing around the figures: the interpretation under each figure, with the numbers that back it up, and the Discussion that closes the report and answers its research question. The lab has three parts. First, compute the numbers behind a figure and write its interpretation; second, write a Discussion for a question that three figures answer together; third, do the same two things for your own report. Parts 1 and 2 use one dataset, the daily weather at Nashville's airport, so that everyone in the section works on the same figures. Part 3 uses yours.

about fifteen minutes to read, and the rest of the section for the sheet

What to download and where to put it

Three files are on Brightspace for this lab:

- this handout;
- the starter sheet, lab-04-interpretations-and-discussion.Rmd/.qmd;
- a data file, nashville_daily.rds.

Please make a folder for the lab inside the course folder (labs/lab-04/ for instance), put the .Rmd/.qmd in it, and put nashville_daily.rds in a data/ folder beside the .Rmd/.qmd.

Part 3 reads your own dataset, so please copy your data file into the same data/ folder, and have your Lab 3 sheet (or your report draft) open: the figures you finished there are the ones Part 3 works on.

What to hand in and when

Rename the starter to Lab-4_FIRSTNAME_LASTNAME.Rmd/.qmd, work through it in section, knit it to PDF, and submit both files on Brightspace before you leave, together with the two data files the sheet reads.

NOTE — How this lab is graded

The syllabus rubric applies, since this lab runs in section. Labs are out of 5 points: one for arriving before the section starts, three for progress (at least half of the sheet by the end of the section), and one for the TA checking your progress when you hand it in. Half of this worksheet is Parts 1 and 2. If you miss your whole lab section in person, then submitting a fully completed worksheet within the week means that you get three progress points and miss the two attendance points.

HINT

The syllabus's fallback applies: knit to HTML, convert the HTML to a PDF with an online converter such as <https://www.sejda.com/html-to-pdf>, and submit that. Please also tell your TA, so that the LaTeX install is sorted out before Report 1 is due.

nashville_daily.rds is the file attached to the 22 September lecture: one row per day at Nashville International Airport, from 1 January 1991 to 17 September 2026. If you already have it, a copy in this lab's data/ folder is all you need.

Part 1: The numbers behind a figure

Lab 3 gave the paragraph that accompanies a figure three jobs: introduce the figure, direct the reader to the feature that matters, and interpret what that feature means for the question. This part is about the last two, and about the numbers they need. “Scores are higher in the review group” directs the reader to a feature. “The median score is 81 in the review group and 74 without it, 7 points higher” directs the reader to the same feature and says how large it is, in the variable's own units, compared to something. A reader can check the second sentence against the figure and repeat it to someone else. (The scores in that example are invented.)

- The worked example: Nashville's summers
- The three figures below answer one research question.
Are Nashville's summers getting hotter, and is it the whole summer or only its hottest days?

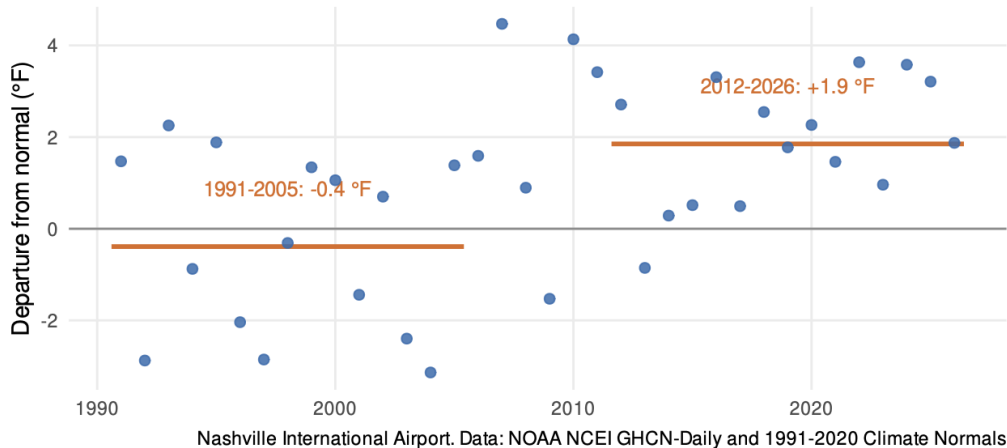
The data are daily high temperatures at Nashville International Airport from the National Oceanic and Atmospheric Administration (NOAA), for June to August of every year from 1991 to 2026. Each figure compares two periods of fifteen summers, 1991–2005 and 2012–2026. Figure 1 gives the level of each summer, Figure 2 the shape

“At the heart of quantitative reasoning is a single question: Compared to what?” Edward Tufte, *Envisioning Information* (1990), p. 67, opening his chapter on small multiples.

of the daily highs in each period, and Figure 3 how often the hottest days came. Each shows something others cannot, which is the test Lab 3 set for the figures in a report.

Nashville's summers were 2.2 °F warmer in 2012-2026 than in 1991-2005

Mean June-to-August daily high minus the 1991-2020 normal; orange: each period's mean



WARNING

"Figure 2 clearly demonstrates a significant upward shift in temperatures over time, highlighting the profound impact of climate change on Nashville's summers."

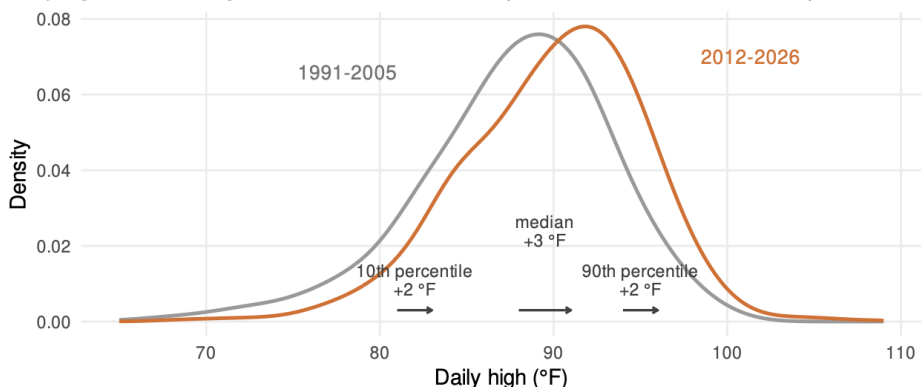
An interpretation, and what is wrong with it

Here is an interpretation of Figure 2 of a kind that often turns up in reports:

It has five problems. First, it has no number: the reader is not told how far the curve moved. Second, "significant" is a technical word. It means that a test was run and its p value was small, and no test was run here. Third, "over time" describes a trend, and the figure compares two periods; it cannot show whether the change was gradual. Fourth, "the impact of climate change" is a claim about a cause, and a figure of observed temperatures

The whole summer moved warmer, not only its hottest days

Daily highs, June to August, fifteen summers in each period; arrows: the shift of three percentiles



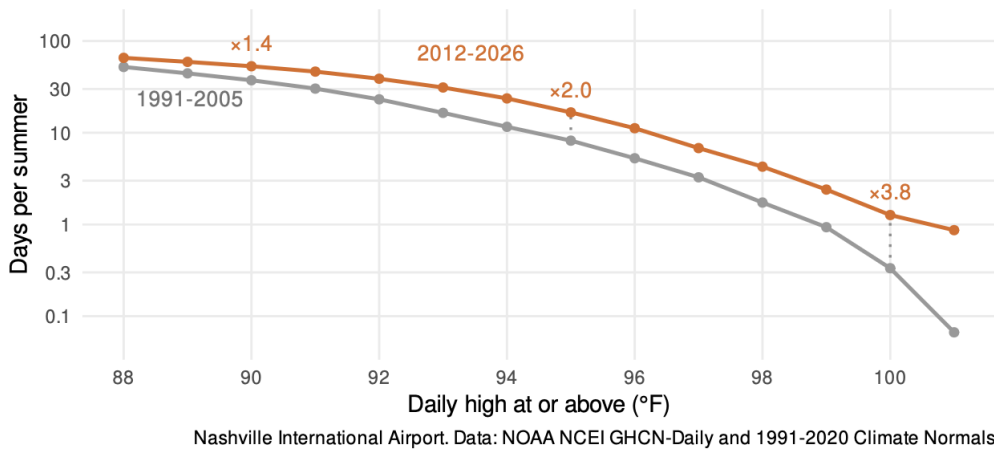
These are Figures 2a, 2b and 2c on the [24 September lecture page](#), where the code that draws each one is given.

Figure 1: The mean daily high of each summer (June to August), 1991 to 2026, as a departure from the 1991-2020 normal, one point per summer. The orange lines are the means of the two periods.

Figure 2: The daily highs of every summer day in each period (1,380 days in each) as density curves. The arrows mark the shift of the 10th percentile, the median, and the 90th percentile.

The hotter the day, the more it has multiplied: 3.8 times as many days at 100 °F

Days per summer (June to August) with a high at or above each temperature; log scale



shows what happened, not why. Fifth, “clearly demonstrates” and “profound” are adjectives doing the work a number should do. The sentence does not identify a specific feature of this figure or provide evidence for its claims.

HINT

“Figure 2 shows the daily highs of every summer day in the two periods, 1,380 days in each. The whole curve sits further to the right in 2012–2026: the median day was 91 °F, against 88 °F in 1991–2005, and the 10th and 90th percentiles each rose by 2 °F. So the warming is not confined to the hottest days; ordinary days warmed about as much. The figure pools fifteen summers into each curve, so it cannot show whether the change was steady from year to year (Figure 1 shows the summers one by one), and days at 100 °F are too rare to show on a density curve at all (Figure 3 counts them).”

Figure 3: Days per summer with a high at or above each temperature from 88 to 101 °F, in each period, on a log scale.

The better version does four things, in this order:

1. It names the comparison. 2012–2026 against 1991–2005. A number with nothing to compare it to cannot be judged large or small.
2. It gives the number, in its units. The median of each period in °F, and the change. The reader can find both on the figure.
3. It says what the number means for the question. The question asked whether it is the whole summer or only its hottest days, and the paragraph answers that part of it.
4. It says what the figure cannot establish. Explain which part of the question this figure cannot answer and, where relevant, which other figure addresses it. This helps keep the interpretation within the evidence shown.

Words that claim more than a figure shows

Four kinds of word appear often in weak interpretations, and each has a plainer replacement that says what the figure shows.

In place of	Write	Because
significant, significantly	the size of the difference, in its units ("2.2 °F warmer")	"significant" says a test was run, and Report 1 does not need one
over time, a trend	the two things compared ("between 1991–2005 and 2012–2026")	only a figure with time on an axis shows a trend
causes, drives, leads to, the impact of	"is higher among", "goes with", "is associated with"	an observational dataset shows what occurred together, not why (Lab 2, Part 3)
clearly, proves, demonstrates, dramatic	"shows", and the number	a number specifies the size of the difference

Computing the numbers, not typing them

Every number in an interpretation should come from the data by code, as the numbers in Lab 2's variables of interest did: inline R in the sentence, so that the number in the text and the figure are drawn from the same rows. The numbers an interpretation needs are usually few, and they come from functions you have already used:

R CODE

```
summers <- nashville |>
  filter(month \%in%\% 6:8) |>
  mutate(period = case_when(
    year <= 2005 ~ "1991-2005",
    year >= 2012 ~ "2012-2026")) |>
  filter(!is.na(period))
early <- filter(summers, period == "1991-2005")
late <- filter(summers, period == "2012-2026")
```

With those two tibbles, `median(early$tmx_f)` is the median summer day of the first period, `quantile(late$tmx_f, 0.9)` is the 90th percentile of the second, and `mean(late$tmx_f >= 95)` is the share of its days at 95 °F or more. (The mean of a logical vector is the proportion of TRUE, since R counts TRUE as 1.) In the sentence, written in the text rather than in a chunk:

R CODE

```
the median day was `r median(late$tmx_f)`degrees F, against
`r median(early$tmx_f)`degrees F in 1991-2005
```

Report 1 asks for summaries and differences computed on your data, not for tests or p values. Use "statistically significant" only when reporting a statistical test that supports that conclusion.

Python stub: use pandas filtering and `np.select()` to create the same period variable, then subset `early` and `late`.

Those who will be working with Python can use an equivalent chunk in their .qmd file as the following:

PYTHON STUB

```
the median day was `{python} late["tmax_f"].median()``degrees F, against
`{python} early["tmax_f"].median()``degrees F in 1991-2005
```

Each inline expression stays on one line, as in Lab 2: a line break inside the backticks stops it being evaluated, and the reader sees the code instead of the number. `round(x, 1)` for one decimal place keeps a number to the precision the reader needs.

A small table, when the numbers are many

If a paragraph contains more than three or four numbers, a small table beside the figure may make the comparison easier to read. `group_by()` and `summarize()` build it, and `kable()` from the `\{knitr\}` package prints it as a formatted table in the knitted PDF, with a caption:

R CODE

```
summers |>
  group_by(period) |>
  summarize(
    days = n(),
    median = median(tmax_f),
    p90 = quantile(tmax_f, 0.9),
    share_95 = mean(tmax_f >= 95)) |>
  knitr::kable(digits = 2,
    col.names = c("Period", "Days", "Median (F)",
      "90th percentile (F)",
      "Share of days at 95 F or more"),
    caption = "Daily summer highs at Nashville in two periods.")
```

PYTHON STUB

```
import pandas as pd
summary_table = (
    summers
    .groupby("period")
    .agg(
        Days=("tmax_f", "size"),
        Median_F=("tmax_f", "median"),
        P90_F=("tmax_f", lambda x: x.quantile(0.9)),
        Share_95=("tmax_f", lambda x: (x >= 95).mean())
    )
    .reset_index()
)
summary_table.columns = [
    "Period",
    "Days",
    "Median (F)",
    "90th percentile (F)",
    "Share of days at 95 F or more"
]
```

Follow it up with a separate stub in your .qmd file to display the table.

PYTHON STUB

```

```{python}
#| label: tbl-summer-highs
#| tbl-cap: "Daily summer highs at Nashville in two periods."

summary_table.round(2)
```

```

A table in a report is for the reader, so it follows the rules of a figure for a reader: columns named in words with their units, the rows the comparison needs and no others, and as many decimal places as the reader can use. A printout of `summary()` is not a table for a reader (Lab 1, Part 2).

TASK

TASK 1 — The numbers behind Figures 2 and 3

Part 1 of the sheet reads `nashville_daily.rds`, builds the two periods, and draws Figure 2 again. Please compute the numbers the better interpretation above cites (the two medians and the shift of each percentile) and write that interpretation in your own words, with every number as inline R/Python. Then do the same for Figure 3: compute the days per summer at 95 °F or more in each period (the number of such days divided by the period's fifteen summers) and how many times as common they became, and write two or three sentences that direct the reader to that feature and say what it means for the question. Last, the sheet has three weak sentences about the figures; please rewrite each so that it claims no more than its figure shows.

Part 2: The Discussion

The interpretations say what each figure shows. The Discussion, at the end of the report, says what the figures show together. It answers the research question and explains what the findings mean, including the limits of the evidence. It has four parts, usually a short paragraph each: first, a summary that answers the question; second, the implications; third, the limitations; fourth, the future directions.

Summary: the answer to the question

Two or three sentences that answer the research question in its own terms. The answer should need all three figures. If one figure answers the question on its own, consider revising the question or replacing figures so that each contributes a distinct finding (Lab 3, Part 3).

WARNING

"Figure 1 showed a difference of 2.2 °F. Figure 2 showed that the median rose from 88 °F to 91 °F. Figure 3 showed that days at 100 °F became 3.8 times as common."

HINT

“Nashville’s summers were 2.2 °F warmer in 2012–2026 than in 1991–2005, and the change was in the whole summer rather than only its hottest days: every part of the distribution of daily highs moved up by two to three degrees. That modest shift multiplied the hottest days the most. Days at 95 °F doubled, from about 8 a summer to about 17, and days at 100 °F went from one summer in three to more than one a summer.”

The first example lists findings from the three figures without explaining how they answer the question together. The summary answers both parts of the question (are summers hotter, and is it the whole summer or the hot days?) and states the conclusion that none of the three figures reaches alone: the higher daily temperatures were accompanied by a large proportional increase in the frequency of very hot days.

Implications: for whom, and what they would do

Explain who could use the finding and how it might inform what they do. Keep the proposed action consistent with the evidence available.

HINT

“Anyone who plans for summer heat in Nashville from the records of the 1990s (a school scheduling outdoor practice, say, or a utility estimating its peak demand) would expect about half as many days at 95 °F as recent summers have had.”

WARNING

“These findings have important implications for climate policy.” This needs more detail: who could use the finding, and for what decision?

The first implication claims only what the figures show: a count of days in the recent past. It does not say that the next fifteen summers will be like the last fifteen, which these data cannot show. Report 1 describes the data you have, and every number in it is a fact about those rows: these 2,760 summer days, or the students who answered a survey. Whether a pattern holds beyond the rows (for the summers to come, or for students who did not answer) is a question about a population, and the Discussion is the place to say how far you think the answer reaches and why.

Limitations: what is specific to your data

Every dataset and every analysis has limitations, and naming them is part of the analysis. The useful ones are specific to your data and your question. There are four places to look: the sample (who or what is in the data, and who or what is missing), the measurement (how each variable was recorded, and whether it measures what the question is about), the design (an observational dataset shows what occurred together, not why), and the scope (the time, the place, and the range of values the data cover).

HINT

“The data come from one station, at the airport, and the ground around an airport can change a great deal in thirty-six years (new runways and parking, and the city growing toward it). The figures cannot separate a warming of the region from a warming of the ground around one thermometer. And the comparison uses the daily highs only: how hot a day feels also depends on the humidity, and on how far it cools at night, which none of the three figures shows.”

WARNING

“More data would make the results more robust.” “There could be confounding variables.” Every dataset could be larger, and every observational dataset could have confounders. A limitation is worth writing when it names what, in this dataset, might have produced the pattern or hidden one.

As in Lab 2, simply stating that your chosen dataset is synthetic or small does not meet the requirement. Discuss limitations specific to the sample, measurements, design, or scope, and explain how they affect the conclusions.

Future directions: the next analysis

What would you do next, with more time, more data, or a method we have not met yet? The best future directions follow from the limitations, and they are specific enough that someone else could carry them out.

HINT

“Drawing the same three figures for a rural station near Nashville would show whether the airport warmed more than its surroundings. Adding the humidity, which NOAA’s hourly records for the same airport include, would turn each daily high into a heat index, which is closer to the heat a person feels.”

WARNING

“Future research could collect more data and use more advanced models.” More data of what kind, and a model for what purpose?

NOTE — A pattern found by looking

An exploratory report looks at the data first, and its research question often sharpens as you explore (Lab 3’s advice). A pattern you found by looking is worth reporting. If you developed or revised the question after seeing a pattern, say so in the Discussion. A useful next step is to check whether the pattern also appears in data you have not yet examined.

TASK

TASK 2 — A Discussion for Nashville’s summers

Part 2 of the sheet has a place for each of the four parts. Please write a Discussion for the research question above, in your own words and with your own examples: first, a summary that needs all three figures; second, one implication matched to its evidence; third, two limitations specific to this dataset, different from the ones above; fourth, one future direction that follows from one of your limitations. Between 150 and 250 words in all is plenty.

Part 3: Your report

Parts 1 and 2 were practice on one question with three finished figures. Part 3 is the same work on your own report, and what you write here is a draft of its interpretations and its Discussion.

TASK

TASK 3 — The numbers behind your figures

Please bring your figures from Lab 3 (or their current versions) into Part 3 of the sheet, one chunk each. Under each figure, compute the numbers its paragraph needs in a chunk of its own (with `summarize()`, or the functions from Part 1), then rewrite the paragraph: introduce the figure by its number, direct the reader to the feature that matters with the number that shows it (in its units and compared to something), say what it means for your question, and say what the figure cannot establish. Every number in the text is inline R. Please check each paragraph against the table of words that claim more than a figure shows.

TASK

TASK 4 — Your Discussion

Please draft the four parts of your report's Discussion under the last heading of the sheet, and write the summary first. If you cannot write an answer that uses a distinct finding from each figure, consider replacing a figure or refining the question. The aim is for all three figures to contribute to the answer.

Five common mistakes

1. A figure with no interpretation. A chunk draws a figure and the next thing the reader sees is another chunk. Every figure is followed by its own paragraph (Lab 3, Part 3).
2. A description of the chart instead of the data. "This boxplot shows the distribution of GPA by major" says what kind of figure it is. Which major's box sits highest, and by how much?
3. A number with no comparison. "The median is 3.4" needs context, such as the median of a comparison group.

4. Words that claim more than the figure shows. A difference called “significant” with no test run, a cause read off an observational dataset, or a trend read off two periods.
5. A Discussion that is a second Results section. It lists the numbers again, figure by figure, and its limitations could have been written about any dataset.

When you have finished

Knit the sheet and check three things in the PDF: each paragraph names its figure by its number; every number in the text is a number and not a line of code (a stray backtick means an inline expression was broken across a line); and the Discussion’s summary needs all three figures. Then show your TA where you got to, and upload both files and the data to Brightspace before you leave.

READING — REFERENCE

- Lab 3, Part 3, for the paragraph that accompanies a figure (introduce, direct, interpret) and for captions; Lab 2, Part 3, for the research question and why a report on observed data does not ask whether X causes Y; Lab 1, Part 2, for the sections of a report and what not to print.
- The Report 1 brief and rubric, on Brightspace.
- Yihui Xie, Christophe Dervieux, and Emily Riederer, *R Markdown Cookbook*, free online: section 10.1, The function `kable()`.
- Claus Wilke, *Fundamentals of Data Visualization*, free online: Chapter 22, Titles, captions, and tables.

SUBMISSION

- ☐ Knit your document to PDF
- ☐ Check that all code is visible (not running off the page)
- ☐ Check that all output displays correctly
- ☐ Submit both `.Rmd/.qmd` and `.pdf` files